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Inventor(s)

Yun; Sun Woong et al.

INTEGRATED CIRCUIT (IC) PACKAGES EMPLOYING STIFFENER STRUCTURE WITH INTEGRATED CAVITY(IES) TO ENCLOSE AN ELECTRICAL COMPONENT(S), AND RELATED FABRICATION METHODS

Abstract

Integrated circuit (IC) packages employing a stiffener structure substrate with integrated cavity(ies) to enclose an electrical component(s), and related fabrication methods are disclosed. The IC package includes a stiffener structure coupled to a first side of the package substrate to increase stiffness of the package substrate, which in turn increases stiffness of the IC package to avoid/reduce warpage. The stiffener structure includes an integrated cavity(ies) extending to the first side of the package substrate and encompasses an electrical component coupled to the first side of the package substrate. In this manner, the stiffener structure does not have to be fully displaced outside of the electrical component to provide room for both the stiffener structure and the electrical component on the package substrate. This area savings can be used to increase stiffener structure size for increased stiffness and without increasing IC package size, or reduce the IC package size.

Inventors: Yun; Sun Woong (San Diego, CA), Buot; Joan Rey Villarba (Escondido, CA), Wang; Zhijie (San Diego, CA)

Applicant: QUALCOMM Incorporated (San Diego, CA)

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Background/Summary

BACKGROUND

I. Field of the Disclosure

[0001] The field of the disclosure relates to integrated circuit (IC) packages, and more particularly to adding stiffener structures to the IC package to increase stiffness and in turn reduce or avoid warpage.

II. Background

[0002] Integrated circuits (ICs) are the cornerstone of electronic devices. ICs are typically packaged in an IC package, also called a “semiconductor package” or “chip package.” The IC package includes a package substrate and one or more semiconductor dies (“dies”) (e.g., in the form of a chip) or other electronic modules mounted to the package substrate to provide electrical connectivity to the die(s). For example, a die in an IC package may be a system-on-a chip (SoC). The dies are electrically coupled to another die(s) and/or to other components in the IC package through electrical coupling to metal lines in the package substrate. The dies can also be electrically coupled to other circuits outside the IC package through electrical connections of external metal interconnects (e.g., solder bumps) of the IC package. The IC package may also include passive electrical components coupled to the package substrate outside of the die(s), but electrically coupled to a die(s) to form part of the circuits of the IC die. These passive components may not be included within the IC die due to size limitations. For example, such a passive electrical component could be a capacitor that is used to provide a decoupling capacitance to shunt noise in power signals to reduce its effect on the IC die(s). Another example of such a passive electrical component is an inductor.

[0003] Providing an IC package with multiple components such as dies and passive electrical components increases the size of the IC package. The package substrate must be large enough to support the multiple components being mounted on the package substrate. The components may also require adjacent keep-out-zones (KoZ) to provide benefits such as providing buffer areas to prevent interference or damage from neighboring components such as from their underfill material, thermal management, and/or reduction in signal interference. However, an increased IC package size can make the IC package more susceptible to warpage from a force applied to the IC package, which in turn can lead to solder joint cracking or failure between the components and the package substrate.

SUMMARY OF THE DISCLOSURE

[0004] Aspects disclosed herein include integrated circuit (IC) packages employing a stiffener structure with an integrated cavity(ies) to enclose an electrical component(s). Related fabrication methods are also disclosed. The IC package includes a package substrate that extends in a first, lateral direction and one or more semiconductor dies (“dies”) coupled to a first side of the package substrate. The IC package also includes one or more electrical components (e.g., a passive electrical component, another die or IC chip) coupled to the first side of the package substrate. The IC package also includes a stiffener structure that is coupled to the first side of the package substrate and that extends in a second, vertical direction from the package substrate. A stiffener

structure is a structure made from one or more materials with higher mechanical strength, such as a metal or composite material that can be coupled or attached to a package substrate to add and increase the stiffness of the package substrate, which in turn increases the stiffness of the IC package to avoid or reduce warpage. In exemplary aspects, the stiffener structure includes one or more integrated cavities that each extend to the first side of the package substrate. The stiffener structure is coupled to the first side of the package substrate such that the cavity encompasses an electrical component coupled to the first side of the package substrate. In an example, the cavity and the first side of the package substrate fully encompass the electrical component. In this manner, the stiffener structure does not have to be fully displaced outside of the electrical component in the first, lateral direction, to provide room for both the stiffener structure and the electrical component. In one example, this can allow the stiffener structure to be expanded in width in the first, lateral direction to increase the stiffness of the IC package without having to enlarge the size of the package substrate and IC package in the first, lateral direction. In another example, this can allow the package substrate to be reduced in width by relocating the stiffener structure to at least partially intersect a vertical area of the electrical component, thereby reducing the size of the IC package.

[0005] In another exemplary aspect, encompassing an electrical component in the cavity of the stiffener structure can also allow the electrical component to be more closely located to a die in the first lateral direction thereby reducing signal routing distance and reducing signal resistance and/or inductance. This is because an outer side wall of the stiffener structure between its cavity and the other electrically-coupled die can act as an underfill material shield or dam to prevent an underfill material disposed around the die from interfering with the electrical component encompassed in the cavity. Thus, a larger lateral keep-out-zone (KoZ) is not required between the electrical component and the die coupled to the package substrate, thus allowing the electrical component to be coupled closer to the die in the first, lateral direction. In an exemplary fabrication method, the stiffener structure is coupled to the package substrate such that its cavity encompasses the electrical component, before an underfill material is disposed around the die. The underfill material is then disposed around the die. In this manner, the outer side wall of the stiffener structure present between its cavity and the other electrically-coupled die provides the underfill material shield or dam to prevent or reduce the underfill material from interfering with the electrical component.

[0006] In another exemplary aspect, the stiffener structure can be provided in the form of a stiffener ring that is disposed in a closed structure around or adjacent to a perimeter of the package substrate and coupled to the package substrate. The stiffener ring is a ring-shaped stiffener structure that is coupled to the first, outer side of the package substrate and extends from the package substrate in the second, vertical direction. One or more cavities are disposed in the ring-shaped stiffener structure that can encompass an electrical component(s) when coupled to the package substrate. As discussed above, an outer side wall of the stiffener ring between its cavity(ies) and the other electrically-coupled die can act as a underfill material shield or dam to prevent an underfill material disposed around the die from interfering with the electrical component encompassed in the cavity, allowing a reduced lateral distance between the encompassed electrical component and the electrically-coupled die to reduce signal routing distance. In another exemplary aspect, the stiffener structure can be an open structure that is not in the form of a ring structure, but rather includes one or more structures that are coupled to the package substrate to add stiffness, and any of which that can include a cavity(ies) to encompass an electrical component(s).

[0007] In another exemplary aspect, the stiffener structure is a heat spreader that includes a lid or cover structure located adjacent to one or more dies coupled to the package substrate. The heat spreader includes a stiffener structure(s) that extend down from the lid or cover structure as a monolithic structure (e.g., created from a mold) and are coupled to the package substrate to support the lid or cover structure being disposed adjacent to the die(s) to dissipate heat generated from the die(s). In this regard, one or more of the stiffener structure(s) of the heat spreader include an integrated cavity that encompasses an electrical component(s) coupled to the package substrate.

[0008] In another exemplary aspect, the stiffener structure can be provided as part of a heat spreader coupled to the package substrate and a separate lid or cover structure coupled to the stiffener structure. This is opposed to a monolithic heat spreader. This form of a heat spreader allows stiffener structure(s) to be coupled to the package substrate with their cavity(ies) encompassing an electrical component(s) on the package substrate first, before the lid or cover structure is coupled to the stiffener structure(s). In this manner, the die area of the package substrate is still open such that an underfill material can then be disposed around a die(s) coupled to the package substrate. The outer side wall of the stiffener structure(s) between its cavity and the die(s) is present to provide the underfill material shield or dam like discussed above. After the underfill material is disposed around the die(s), the lid or cover structure can then be subsequently coupled to the stiffener structure(s) to completely form a heat spreader and encompass the die(s) within the heat spreader.

[0009] In this regard, in one exemplary aspect, an IC package is provided. The IC package comprises a package substrate comprising a first side extending in a first direction. The IC package also comprises a first die coupled to the first side of the package substrate. The IC package also comprises a first stiffener structure adjacent to the first die in the first direction and coupled to the first side of the package substrate. The first stiffener structure comprises a first cavity extending adjacent to the first side of the package substrate. The IC package also comprises a first electrical component coupled to the first side of the package substrate, the first electrical component at least partially embedded in the first cavity.

[0010] In another exemplary aspect, a method of fabricating an IC package is provided. The method comprises providing a package substrate comprising a first side extending in a first direction. The method also comprises coupling a first electrical component to the first side of the package substrate. The method also comprises coupling a first surface of a first stiffener structure comprising a first cavity extending adjacent to the first surface, to the first side of the package substrate such that the first cavity at least partially encloses the first electrical component. The method also comprises coupling a first die to the first side of the package substrate adjacent to the first stiffener structure in the first direction.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0011] FIGS. 1A and 1B are respective side and top perspective views of an exemplary integrated circuit (IC) package that includes multiple semiconductor dies (“dies”) and die-side decoupling capacitors mounted to a package substrate and further includes an outer metal stiffener ring structure to add stiffness to the IC package to reduce or avoid warpage;

[0012] FIGS. 2A and 2B are respective side and top views of an exemplary IC package that includes a stiffener structure in the form of a stiffener ring coupled to a first side of a package substrate to increase stiffness of the IC package, and wherein the stiffener ring includes an integrated closed cavity extending to the first side of the package substrate and encompassing electrical components coupled to the first side of the package substrate;

[0013] FIGS. 3A and 3B are respective side and top views of an exemplary IC package that includes a stiffener structure in the form of a stiffener ring coupled to a first side of a package substrate to increase stiffness of the IC package, and wherein the stiffener ring includes an integrated open cavity extending to the first side of the package substrate and encompassing electrical components coupled to the first side of the package substrate;

[0014] FIGS. 4A and 4B are respective side and top views of another exemplary IC package that includes a stiffener structure in the form of a heat spreader that includes a ring-shaped stiffener structure coupled to a first side of a package substrate to increase stiffness of the IC package and to

support a separate lid structure coupled to the stiffener structure to provide a heat spreader, and wherein the ring-shaped stiffener structure includes an integrated cavity extending to the first side of the package substrate and encompassing electrical components coupled to the first side of the package substrate;

[0015] FIGS. 5A and 5B are respective side and top views of another exemplary IC package that includes a stiffener structure in the form of a heat spreader that includes a ring-shaped stiffener structure coupled to a first side of a package substrate to increase stiffness of the IC package and to support a lid structure coupled to the stiffener as a monolithic structure to provide a heat spreader, and wherein the ring-shaped stiffener structure includes an integrated cavity extending to the first side of the package substrate and encompassing electrical components coupled to the first side of the package substrate;

[0016] FIG. 6 is a flowchart illustrating an exemplary process of fabricating an IC package that includes a stiffener structure coupled to a first side of a package substrate to increase stiffness of the IC package, and wherein the stiffener structure includes an integrated cavity(ies) extending to the first side of the package substrate and encompassing an electrical component(s) coupled the first side of the package substrate, including, but not limited to, the IC packages in FIGS. 2A-5B;

[0017] FIGS. 7A and 7B is a flowchart illustrating another exemplary fabrication process of fabricating the exemplary IC package in FIGS. 2A and 2B that includes a stiffener structure in the form of a stiffener ring coupled to a first side of a package substrate to increase stiffness of the IC package, and wherein the stiffener ring includes an integrated cavity extending to the first side of the package substrate and encompassing electrical components coupled to the first side of the package substrate;

[0018] FIGS. 8A-8C are exemplary fabrication stages during fabrication of the IC package according to the exemplary fabrication process in FIGS. 7A and 7B;

[0019] FIGS. 9A-9C is a flowchart illustrating another exemplary fabrication process of fabricating the exemplary IC package in FIGS. 4A and 4B that includes a stiffener structure in the form of a heat spreader that includes a ring-shaped stiffener structure coupled to a first side of a package substrate to increase stiffness of the IC package and to support a separate lid structure coupled to the stiffener structure to provide a heat spreader, and wherein the ring-shaped stiffener structure includes an integrated cavity extending to the first side of the package substrate and encompassing electrical components coupled to the first side of the package substrate;

[0020] FIGS. 10A-10E are exemplary fabrication stages during fabrication of the IC package according to the exemplary fabrication process in FIGS. 9A-9C;

[0021] FIGS. 11A and 11B is a flowchart illustrating another exemplary fabrication process of fabricating the exemplary IC package in FIGS. 5A and 5B that includes a stiffener structure in the form of a heat spreader that includes a ring-shaped stiffener structure coupled to a first side of a package substrate to increase stiffness of the IC package and to support a lid structure coupled to the stiffener as a monolithic structure to provide a heat spreader, and wherein the ring-shaped stiffener structure includes an integrated cavity extending to the first side of the package substrate and encompassing electrical components coupled to the first side of the package substrate;

[0022] FIGS. 12A-12D are exemplary fabrication stages during fabrication of the IC package according to the exemplary fabrication process in FIGS. 11A and 11B;

[0023] FIG. 13 is a block diagram of an exemplary wireless communications device that includes one or more IC packages that include a stiffener structure coupled to a first side of a package substrate to increase stiffness of the IC package, and wherein the stiffener structure includes an integrated cavity(ies) extending to the first side of the package substrate and encompassing an electrical component(s) coupled to the first side of the package substrate, including, but not limited to, the IC packages in FIGS. 2A-5B, 8C, 10E, and 12D, and that can be fabricated according to a fabrication process, including, but not limited to, the exemplary fabrication processes in FIGS. 6, 7A-7B, 9A-9C, and 11A-11B; and

[0024] FIG. 14 is a block diagram of an exemplary electronic device in the form of a processor-based system that can include one or more IC packages that include a stiffener structure coupled to a first side of a package substrate to increase stiffness of the IC package, and wherein the stiffener structure includes an integrated cavity(ies) extending to the first side of the package substrate and encompassing an electrical component(s) coupled to the first side of the package substrate, including, but not limited to, the IC packages in FIGS. 2A-5B, 8C, 10E, and 12D, and that can be fabricated according to a fabrication process, including, but not limited to, the exemplary fabrication processes in FIGS. 6, 7A-7B, 9A-9C, and 11A-11B.

DETAILED DESCRIPTION

[0025] With reference now to the drawing figures, several exemplary aspects of the present disclosure are described. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects.

[0026] Aspects disclosed herein include integrated circuit (IC) packages employing a stiffener structure with an integrated cavity(ies) to enclose an electrical component(s). Related fabrication methods are also disclosed. The IC package includes a package substrate that extends in a first, lateral direction and one or more semiconductor dies (“dies”) coupled to a first side of the package substrate. The IC package also includes a stiffener structure that is coupled to the first side of the package substrate and that extends in a second, vertical direction from the package substrate. A stiffener structure is a structure made from one or more materials with higher mechanical strength, such as a metal or composite material that can be coupled or attached to a package substrate to add and increase the stiffness of the package substrate, which in turn increases the stiffness of the IC package to avoid or reduce warpage. In exemplary aspects, the stiffener structure includes one or more integrated cavities that each extend to the first side of the package substrate. The stiffener structure is coupled to the first side of the package substrate such that the cavity encompasses an electrical component coupled to the first side of the package substrate. In an example, the cavity and the first side of the package substrate fully encompass the electrical component. In this manner, the stiffener structure does not have to be fully displaced outside of the electrical component in the first, lateral direction, to provide room for both the stiffener structure and the electrical component. In one example, this can allow the stiffener structure to be expanded in width in the first, lateral direction to increase the stiffness of the IC package without having to enlarge the size of the package substrate and IC package in the first, lateral direction. In another example, this can allow the package substrate to be reduced in width by relocating the stiffener structure to at least partially intersect a vertical area of the electrical component, thereby reducing the size of the IC package.

[0027] Before discussing examples of IC packages that include a stiffener structure coupled to a first side of a package substrate to increase stiffness of the IC package, and wherein the stiffener structure includes an integrated cavity(ies) extending to the first side of the package substrate and encompassing an electrical component(s) coupled to the first side of the package substrate starting at FIG. 2A, an example of an IC package that does not include a stiffener structure encompassing an electrical component is first discussed with regard to FIGS. 1A and 1B.

[0028] In this regard, FIGS. 1A and 1B are respective side and top perspective views of an exemplary integrated circuit (IC) package 100 that includes multiple semiconductor dies (“dies”) 102(1), 102(2) and die-side capacitors 104(1), 104(2) mounted to a package substrate 106. The side view of the IC package 100 in FIG. 1A is a cross-sectional view across the A.sub.1-A.sub.1' cross-section line in the IC package 100 in FIG. 1B. The capacitors 104(1), 104(2) are examples of electrical components. The capacitors 104(1), 104(2) may be coupled to circuits in the dies 102(1) and/or 102(2) through signal routing paths through the package substrate 106 to shunt noise from one electrical circuit (e.g., a power supply circuit) to another electrical circuit (e.g., a powered electrical circuit). As another example, the capacitors 104(1), 104(2) may form part of a filtering circuit in the dies 102(1) and/or 102(2). The IC package 100 also includes an outer metal ring

stiffener structure **108** to add stiffness to the package substrate **106** to add stiffness to the IC package **100** to reduce or avoid warpage.

[0029] As shown in FIG. 1A, the capacitors **104(1)**, **104(2)** are laterally displayed in a first, horizontal direction (X-axis, Y-axis directions) from respective side walls **110(1)**, **110(2)** of the dies **102(1)**, **102(2)** so that the capacitors **104(1)**, **104(2)** are outside respective keep-out-zones KOZ.sub.1, KOZ.sub.2 for the dies **102(1)**, **102(2)**. The keep-out-zones KOZ.sub.1, KOZ.sub.2 in this example are areas that are designed to remain free from interference from other components so that an underfill material **112** can be disposed underneath and adjacent to the dies **102(1)**, **102(2)** to reinforce their stability and their solder joints to the package substrate **106**. However, providing these keep-out-zones KOZ.sub.1, KOZ.sub.2 increases the lateral width of the package substrate **106** in the first, horizontal direction (X-axis and/or Y-axis directions), because the capacitors **104(1)**, **104(2)** and the stiffener structure **108** must be laterally displaced outside of the respective keep-out-zones KOZ.sub.1, KOZ.sub.2 in the first, horizontal direction (X-axis and/or Y-axis directions). This leads to an increase in the size of the package substrate **106**, which in turn leads to an increase in the size of the overall IC package **100**, which may be undesirable, especially for mobile device applications where minimizing IC package size is important. The stiffener structure **108** could be reduced in size or width in the first, horizontal direction (X-axis and/or Y-axis directions) to allow for a smaller size package substrate **106** in the IC package **100**. However, this may reduce the added stiffness to the package substrate **106** making it and the IC package **100** more susceptible to warpage and resulting damage in an undesired manner.

[0030] In this regard, FIGS. 2A and 2B are respective side and top views of an exemplary IC package **200** that includes first and second dies **202(1)**, **202(2)** and a stiffener structure **208** that has first and second cavities **212(1)**, **212(2)** for encompassing respective electrical component(s) **204(1)**, **204(2)** coupled to a package substrate **206**. As discussed in more detail below, the stiffener structure **208** is a structure made from one or more materials with higher mechanical strength, such as a metal or composite material that can be coupled or attached to the package substrate **206** to add and increase the stiffness of the package substrate **206**, which in turn increases the stiffness of the IC package **200** to avoid or reduce warpage. The stiffener structure **208** and its first and second cavities **212(1)**, **212(2)** have respective vertical axes VA.sub.1, VA.sub.2 that intersect the respective electrical components **204(1)**, **204(2)** in a second, vertical direction (Z-axis direction) orthogonal to the first, horizontal direction (X-axis and/or Y-axis directions). The stiffener structure **208** is located outside the first and second dies **202(1)**, **202(2)** in the first, horizontal direction (X-axis and/or Y-axis directions) and neither it nor its first and second cavities **212(1)**, **212(2)** intersects either the first or second die **202(1)**, **202(2)** in the second, vertical direction (Z-axis direction). In this example, as shown in FIG. 2B, the stiffener structure **208** is a single structure that is a ring-shaped stiffener structure that is adjacent to the perimeter **216** of the package substrate **206**, but such is not required. The stiffener structure **208** could be a structure segment that is not a closed shaped or multiple structures that each include the respective first and second cavities **212(1)**, **212(2)** encompassing the respective first and second electrical components **204(1)**, **204(2)**. As discussed in more detail below, this provides additional area for the stiffener structure **208** and electrical components **204(1)**, **204(2)** to not have to be fully displaced from each other in the first, horizontal direction (X-axis and/or Y-axis directions) such that the package substrate **206** has to be enlarged in width in the first, horizontal direction (X-axis and/or Y-axis directions) to support both the stiffener structure **208** and the electrical components **204(1)**, **204(2)**.

[0031] In this manner, in one example, the package substrate **206** can be reduced in width W.sub.1 in the first, horizontal direction (X-axis and/or Y-axis directions) with the area saved by the electrical components **204(1)**, **204(2)** being embedded in the first and second cavities **212(1)**, **212(2)** of the stiffener structure **208** to provide for a reduced size IC package **200**, while still providing for the electrical components **204(1)**, **204(2)** and stiffener structure **208** to be provided in the IC package **200**. In another example, providing the first and second cavities **212(1)** and/or

212(2) in the stiffener structure **208** that encompass the electrical components **204(1)**, **204(2)** can allow the stiffener structure **208** to be expanded in width in the first, horizontal direction (X-axis and/or Y-axis directions) to increase the stiffness of the IC package **200** without having to enlarge the size of the package substrate **206** and the IC package **200** in the first, horizontal direction (X-axis and/or Y-axis directions) like shown in the IC package **100** in FIGS. **1A** and **1B** for example. [0032] Note that the example IC package **200** in FIGS. **2A** and **2B** includes multiple dies **202(1)**, **202(2)**, but such is not required. The IC package **200** could include any number of dies desired. Also note that the IC package includes multiple cavities **212(1)**, **212(2)**, but such is not required. The IC package **200** could include any number of cavities. Also note that the first and second cavities **212(1)**, **212(2)** could be provided as a single, common cavity that is continuous through the stiffener structure **208** in the form of a channel structure for example. Also note that the stiffener structure **208** shown in FIG. **2A** could be two separate stiffener structures that are provided independently of each other with their separate respective first and second cavities **212(1)**, **212(2)** to encompass the respective first and second electrical components **204(1)**, **204(2)**.

[0033] With reference to FIG. **2A**, in this example, the first and second dies **202(1)**, **202(2)** are coupled to a first side **214(1)** of the package substrate **206**. For example, the first and second dies **202(1)**, **202(2)** can be discrete, modular dies that are chiplets coupled to the package substrate **206** and electrically coupled to each other using die-to-die connections provided through the package substrate **206**. The dies **202(1)**, **202(2)** as chiplets can include circuitry that is specific to a task or functionality, as opposed to, for example, a system-on-a-chip (SoC) that includes multiple systems and functions on a single, monolithic die. As process node continues to scale and die size increases, reducing the cost of yield loss becomes more challenging. Thus, by breaking a large monolithic die into chiplets, such as the dies **202(1)**, **202(2)**, cost due to yield loss is reduced. Chiplets can facilitate an IC package's faster time to market as chiplets allow already proven and tested dies to be integrated with other chiplets in the IC package. Chiplets also allow for dies, such as the dies **202(1)**, **202(2)**, from different manufacturers to be integrated together in the same package for heterogeneous integration, such as in the IC package **200**. Using chiplets in an IC package, including the IC package **200**, also facilitates the ability to use fabrication technologies and processes of choice to fabricate the first and second dies **202(1)**, **202(2)** as either from the same or different technology nodes.

[0034] For example, the first die **202(1)** as a chiplet may be fabricated using a first technology node, and the second die **202(2)** as a chiplet may be fabricated using a second technology node that is not as advanced as the first technology node. In such an example, the first die **202(1)** may include components (e.g., interconnects, transistors) that have a first minimum size, and the second die **202(2)** may include components (e.g., interconnects, transistors) that have a second minimum size, where the second minimum size is greater than the first minimum size. In some implementations, the first and second dies **202(1)**, **202(2)** may be fabricated using the same technology node or different technology nodes.

[0035] Note that the IC package **200** in FIG. **2A** is not limited to including separate first and second dies **202(1)**, **202(2)**, and as chiplets. The IC package **200** could include a single monolithic die, such as a SoC for example. The IC package **200** could also be a multi-chip module (MCM) that includes IC chips or a flip chip ball grid array (FPBGA) that utilizes a controlled collapse chip connection technology, also known as flip chip, for die **202(1)** or **202(2)** to package substrate **206** interconnection.

[0036] The package substrate **206** may be a laminate type substrate that includes multiple metallization layers laminated to each other each with metal traces to provide signal routing paths. For example, the first and/or second electrical components **204(1)**, **204(2)** may be electrically coupled to the first and/or second dies **202(1)**, **202(2)** through signal routing paths in the package substrate **206**. As another example, the package substrate **206** could be an embedded trace substrate (ETS) that includes embedded metal traces in an insulating layer of one or more metallization

layers that are fabricated on top of each other with vias providing interconnections between the metal traces. The first and second electrical components **204(1)**, **204(2)** are also coupled to the first side **214(1)** of the package substrate **206** laterally adjacent to the respective first and second dies **202(1)**, **202(2)** in the first, horizontal direction (X-axis and/or Y-axis directions). The stiffener structure **208** is also coupled to the first side **214(1)** of the package substrate **206** to add stiffness to the package substrate **206**. For example, the stiffener structure **208** may be a metal frame. An adhesive material **218** may be disposed between a first, outer surface **222** of the stiffener structure **208** and the first side **214(1)** of the package substrate **206** to secure the stiffener structure **208** to the package substrate **206**. In this manner, the first, outer surface **222** of the stiffener structure **208** is still coupled to the first side **214(1)** of the package substrate **206**, but not directly in contact with the first side **214(1)** of the package substrate **206**. The stiffener structure **208** could also be made from a non-metal material that has an increased stiffness to add stiffness to the package substrate **206**. The stiffener structure **208** is disposed on the first side **214(1)** of the package substrate **206** also adjacent to the first and second dies **202(1)**, **202(2)**.

[0037] As shown in FIG. 2A, the stiffener structure **208** includes the first and second cavities **212(1)**, **212(2)** that encompass the respective electrical components **204(1)**, **204(2)**. The electrical components **204(1)**, **204(2)** can be any type of electrical component, including a passive electrical component (e.g., a resistor, capacitor, inductor), or active electrical component that is powered. In this example, the electrical components **204(1)**, **204(2)** are fully embedded in the cavities **212(1)**, **212(2)** between respective upper surfaces **220(1)**, **220(2)** of the stiffener structure **208** and the first side **214(1)** of the package substrate **206**. Note that in other examples discussed below, the electrical components **204(1)**, **204(2)** are partially embedded in the cavities **212(1)**, **212(2)**. The upper surfaces **220(1)**, **220(2)** are formed between the first, outer surface **222** of the stiffener structure **208** and a second, outer surface **224** of the stiffener structure **208**. The first and second cavities **212(1)**, **212(2)** are formed from a void of material in a portion of the stiffener structure **208** between the first, outer surface **222** and between a first, outer side wall **226(1)** and a second, outer side wall **226(2)** of the stiffener structure **208** in the first, horizontal direction (X-axis and/or Y-axis directions). The first and second cavities **212(1)**, **212(2)** extend down to or adjacent to the second, outer surface **224** such that the first and second cavities **212(1)**, **212(2)** have respective openings **228(1)**, **228(2)** open to the first, outer surface **222** of the stiffener structure **208**. In this manner, the first and second cavities **212(1)**, **212(2)** are closed cavities closed in by the respective first and second upper surfaces **220(1)**, **220(2)** of the stiffener structure **208**. The openings **228(1)**, **228(2)** of the first and second cavities **212(1)**, **212(2)** extend down to or adjacent to the first side **214(1)** of the package substrate **206** with the stiffener structure **208** mounted to the first side **214(1)** of the package substrate **206**.

[0038] Further, as shown in FIGS. 2A and 2B, an underfill material **230** is disposed adjacent to and underneath the first and second dies **202(1)**, **202(2)** to reinforce the stability of the first and second dies **202(1)**, **202(2)** and their solder joints **232(1)**, **232(2)** to the package substrate **206**. Another non-limiting benefit of providing the cavities **212(1)**, **212(2)** in the stiffener structure **208** to encompass the first and second electrical components **204(1)**, **204(2)** is that the stiffener structure **208** provides an underfill shield or dam between the underfill material **230** and the first and second electrical components **204(1)**, **204(2)**. In this example, by the stiffener structure **208** being located adjacent to the first and second dies **202(1)**, **202(2)** in the first, horizontal direction (X-axis and/or Y-axis directions), the first, outer side wall **226(1)** of stiffener structure **208** located adjacent to the first and second dies **202(1)**, **202(2)** in the first, horizontal direction (X-axis and/or Y-axis directions) can act as an underfill shield or dam. If the stiffener structure **208** is coupled to the package substrate **206** before the underfill material **230** is disposed around and underneath the first and second dies **202(1)**, **202(2)**, the first, outer side wall **226(1)** provides a shield or dam to prevent the underfill material **230** from reaching and contacting the first and second electrical components **204(1)**, **204(2)**. The underfill material **230** is permitted to contact the first, outer side wall **226(1)**

without issue since the stiffener structure **208** does not provide electrical functionality in the IC package **200**. In this manner, any keep-out-zones KoZ.sub.3, KoZ.sub.4 that are provided between the respective first and second dies **202(1)**, **202(2)** and the stiffener structure **208** may be able to be reduced in width in the first, horizontal direction (X-axis and/or Y-axis directions). This allows first and second distances D.sub.1, D.sub.2 of the first and second electrical components **204(1)**, **204(2)** to respective side walls **234(1)**, **234(2)** of the first and second dies **202(1)**, **202(2)** in the first, horizontal direction (X-axis and/or Y-axis directions) to be reduced, which also allows the width W.sub.1 of the package substrate **206** to be reduced while still being able to support the first and second electrical components **204(1)**, **204(2)** and the stiffener structure **208**. For example, the first second distances D.sub.1, D.sub.2 may be between 0.65 to 1.2 millimeters (mm).

[0039] As discussed in FIGS. 2A and 2B above, the IC package **200** therein includes the stiffener structure **208** that has closed cavities **212(1)**, **212(2)** that are bounded by the respective first and second upper surfaces **220(1)**, **220(2)** of the stiffener structure **208**. However, such is not required. Cavities provided in a stiffener structure to encompass an electrical component could be provided as open cavities that allow the electrical component to be exposed in the second, vertical direction (Z-axis direction) from its cavity, and thus partially encompassed by the cavity. This is shown in the exemplary IC package **300** in FIGS. 3A and 3B as another example. FIGS. 3A and 3B are respective side and top views of an exemplary IC package **300** that includes an alternative stiffener structure **308** that is like the stiffener structure **208** in the IC package **200** in FIGS. 2A and 2B. Common elements between the IC package **300** in FIGS. 3A and 3B and the IC package **200** in FIGS. 2A and 2B are shown with common element numbers and thus are not re-described.

However, as shown in FIGS. 3A and 3B, the stiffener structure **308** in the IC package **300** includes first and second cavities **312(1)**, **312(2)** that are not bounded by first and second upper surfaces **220(1)**, **220(2)**, and thus have openings **322(1)**, **322(2)** in the second, outer surface **224**.

[0040] The stiffener structure provided in an IC package, like the IC packages **200**, **300** in FIGS. 2A-3B can also dissipate heat and/or be part of a heat spreader to dissipate heat. In this regard, FIGS. 4A and 4B are respective side and top views of another exemplary IC package **400** that includes a stiffener structure **208** that is like the stiffener structure **208** in the IC package **200** in FIGS. 2A-2B. Common elements between the IC package **400** in FIGS. 4A-4B and the IC package **200** in FIGS. 2A-2B are shown with common element numbers, and thus are not re-described. However, as shown in FIGS. 4A and 4B, a heat spreader structure **408** is provided that includes the stiffener structure **208** and a cover structure **410**. An optional thermal transfer material **402(1)**, **402(2)** (e.g. a thermal paste, thermal grease) is disposed on the first and second dies **202(1)**, **202(2)** to prepare for the cover structure **410** to be disposed thereon to facilitate heat transfer from the first and second dies **202(1)**, **202(2)** to the cover structure **410**. The cover structure **410** in this example, is a separate component, which may be a metal component, and is coupled to the second, outer surface **224** of the stiffener structure **208** to close off a cavity **412** between the dies **202(1)**, **202(2)** as shown in FIG. 4B. The cover structure **410** extends in the first, horizontal direction (X-axis and Y-axis directions) along the second, outer surface **224** of the stiffener structure **208** such that the cover structure **410** intersects the first and second dies **202(1)**, **202(2)** in the second, vertical direction (Z-axis direction). In this manner, the cover structure **410** is adjacent to the first and second dies **202(1)**, **202(2)** to assist in dissipation of heat generated by the first and second dies **202(1)**, **202(2)**.

[0041] FIGS. 5A and 5B are respective side and top views of another exemplary IC package **500** that includes a heat spreader structure **508** that is like the heat spreader structure **408** in the IC package **400** in FIGS. 4A-4B. Common elements between the IC package **500** in FIGS. 5A-5B and the IC package **400** in FIGS. 4A-4B are shown with common element numbers, and thus are not re-described. However, as shown in FIGS. 5A and 5B, the heat spreader structure **508** is provided that includes the stiffener structure **208** and a cover structure **510** as a single monolithic component. For example, the heat spreader structure **508** may have been fabricated from a mold as a one-piece

component. The heat spreader structure **508** be a metal component, and is coupled to the first side **214(1)** of the package substrate **206**. The cover structure **510** extends in the first, horizontal direction (X-axis and Y-axis directions) such that the cover structure **510** intersects the first and second dies **202(1)**, **202(2)** in the second, vertical direction (Z-axis direction). In this manner, the cover structure **510** is adjacent to the first and second dies **202(1)**, **202(2)** to assist in dissipation of heat generated by the first and second dies **202(1)**, **202(2)**.

[0042] Note that with the stiffener structure **208** being provided as a monolithic part of the heat spreader structure **508** in the IC package **500** in FIGS. 5A and 5B, the cavity **412** is not accessible to dispose the underfill material **230** around the dies **202(1)**, **202(2)** once the heat spreader structure **508** with its stiffener structure **208** is coupled to the package substrate **206**.

[0043] FIG. 6 is a flowchart illustrating an exemplary process of fabricating an IC package that includes a stiffener structure coupled to a first side of a package substrate to increase stiffness of the IC package, and wherein the stiffener structure includes an integrated cavity(ies) extending to the first side of the package substrate and encompassing an electrical component(s) coupled to the first side of the package substrate, including, but not limited to, the IC packages **200**, **300**, **400**, **500** in FIGS. 2A-5B. The fabrication process **600** is discussed in reference to the IC package **200** in FIGS. 2A-2B, but the fabrication process **600** is not limited to fabricating the IC package **200** in FIGS. 2A-2B.

[0044] In this regard, a first step in the fabrication process **600** is providing a package substrate **206** comprising a first side **214(1)** extending in a first direction (X-axis and/or Y-axis directions) (block **602** in FIG. 6). A next step in the fabrication process **600** is coupling a first electrical component **204(1)**, **204(2)** to the first side **214(1)** of the package substrate **206** (block **604** in FIG. 6). A next step in the fabrication process **600** is coupling a first surface **222** of a first stiffener structure **208** comprising a first cavity **212(1)**, **212(2)** extending adjacent to first surface **222**, to the first side **214(1)** of the package substrate **206** such that the first cavity **212(1)**, **212(2)** at least partially encloses the first electrical component **204(1)**, **204(2)** (block **606** in FIG. 6). A next step in the fabrication process **600** is coupling a first die **202(1)**, **202(2)** to the first side **214(1)** of the package substrate **206** adjacent to the first stiffener structure **208** in the first direction (X-axis and/or Y-axis directions) (block **608** in FIG. 6).

[0045] An IC package that includes a stiffener structure coupled to a first side of a package substrate to increase stiffness of the IC package, and wherein the stiffener structure includes an integrated cavity(ies) extending to the first side of the package substrate and encompassing an electrical component(s) coupled to the first side of the package substrate, including, but not limited to, the IC packages **200**, **300**, **400**, **500** in FIGS. 2A-5B, can be fabricated in other fabrication processes.

[0046] For example, FIGS. 7A and 7B is a flowchart illustrating an exemplary fabrication process **700** of fabricating the IC package **200** in FIGS. 2A-2B that includes the stiffener structure **208** coupled to the first side **214(1)** of the package substrate **206** to increase stiffness of the IC package **200**, and wherein the stiffener structure **208** includes an integrated cavity(ies) **212(1)**, **212(2)** extending to the first side **214(1)** of the package substrate **206** and encompassing an electrical component(s) **204(1)**, **204(2)** coupled to the first side **214(1)** of the package substrate **206**. FIGS. 8A-8C illustrate exemplary fabrication stages **800A-800C** during fabrication of the IC package **200** according to the fabrication process **700** in FIGS. 7A and 7B. The fabrication process **700** in FIGS. 7A and 7B is discussed below with reference to the fabrication stages **800A-800C** in FIGS. 8A-8C.

[0047] In this regard, as illustrated in fabrication stage **800A** in FIG. 8A, the fabrication process **700** includes providing the package substrate **206** and coupling the electrical components **204(1)**, **204(2)** to the first side **214(1)** of the package substrate **206** (block **702** in FIG. 7A). Then, as illustrated in fabrication stage **800B** in FIG. 8B, the fabrication process **700** includes coupling the stiffener structure **208** on the first side **214(1)** of the package substrate **206** such that the cavities **212(1)**, **212(2)** of the stiffener structure **208** are aligned with the electrical components **204(1)**,

204(2) and encompass the electrical components **204(1)**, **204(2)** (block **704** in FIG. **7A**). As previously discussed, an optional adhesive material **218** may be first disposed on the first side **214(1)** of the package substrate **206** or on the second, outer surface **224** of the stiffener structure **208** to secure the stiffener structure **208** to the package substrate **206**. Then, as illustrated in fabrication stage **800C** in FIG. **8C**, the fabrication process **700** includes placing and coupling the first and second dies **202(1)**, **202(2)** to the first side **214(1)** of the package substrate **206** and disposing the underfill material **230** adjacent to and underneath the dies **202(1)**, **202(2)** to form the IC package **200** (block **706** in FIG. **7B**). Note that as discussed previously, with the stiffener structure **208** coupled to the package substrate **206** before the underfill material **230** is disposed around the dies **202(1)**, **202(2)**, the first, outer side wall **226(1)** of the stiffener structure **208** acts as a shield or dam to prevent the underfill material **230** from interfering with the first and second electrical components **204(1)**, **204(2)**.

[0048] FIGS. **9A-9C** is a flowchart illustrating an exemplary fabrication process **900** of fabricating the IC package **400** in FIGS. **4A-4B** that includes the stiffener structure **208** that is included in a heat spreader structure **408**. FIGS. **10A-10E** illustrate exemplary fabrication stages **1000A-1000E** during fabrication of the IC package **400** according to the fabrication process **900** in FIGS. **9A-9C**. The fabrication process **900** in FIGS. **9A-9C** is discussed below with reference to the fabrication stages **1000A-1000E** in FIGS. **10A-10E**.

[0049] In this regard, the first three fabrication stages **1000A-1000C** in FIGS. **10A-10C** of the fabrication process **900** and their related process steps **902**, **904**, **906** in FIGS. **9A** and **9B** are the same as the first three fabrication stages **800A-800C** in FIGS. **8A-8C** of the fabrication process **700** in FIGS. **7A-7B**, and thus will not be repeated. As discussed previously, with the stiffener structure **208** coupled to the package substrate **206** before the underfill material **230** is disposed around the dies **202(1)**, **202(2)**, the first, outer side wall **226(1)** of the stiffener structure **208** acts as a shield or dam to prevent the underfill material **230** from interfering with the first and second electrical components **204(1)**, **204(2)**. Then, as illustrated in fabrication stage **1000D** in FIG. **10D**, the fabrication process **900** includes providing an optional thermal transfer material **402(1)**, **402(2)** (e.g. a thermal paste, thermal grease) on the first and second dies **202(1)**, **202(2)** to prepare for the cover structure **410** to be disposed thereon and on the first surface **222** of the stiffener structure **208** (block **908** in FIG. **9B**). The thermal transfer material **402(1)**, **402(2)** will provide a good thermal transfer between the dies **202(1)**, **202(2)** and the cover structure **410** of the heat spreader structure **408** to efficiently dissipate heat. Then, as illustrated in fabrication stage **1000E** in FIG. **10E**, the fabrication process **900** includes disposing the cover structure **410** on the first surface **222** of the stiffener structure **208** and on the thermal transfer material **402(1)**, **402(2)** adjacent to the first and second dies **202(1)**, **202(2)** to form the IC package **400** (block **910** in FIG. **9C**).

[0050] FIGS. **11A-11B** is a flowchart illustrating the exemplary fabrication process **1100** of fabricating the IC package **500** in FIGS. **5A-5B** that includes the stiffener structure **208** in a heat spreader structure **508** as a monolithic structure. FIGS. **12A-12D** illustrate exemplary fabrication stages **1200A-1200D** during fabrication of the IC package **500** according to the fabrication process **1100** in FIGS. **11A-11B**. The fabrication process **1100** in FIGS. **11A-11B** is discussed below with reference to the fabrication stages **1200A-1200D** in FIGS. **12A-12D**.

[0051] In this regard, the first fabrication stage **1200A** in FIG. **12A** of the fabrication process **1100** and its process step **1102** in FIG. **11A** is the same as the fabrication stage **800A** in FIG. **8A** of the fabrication process **700** in FIGS. **7A-7B**, and thus will not be repeated.

[0052] As illustrated in fabrication stage **1200B** in FIG. **12B**, the fabrication process **1100** then includes placing and coupling the first and second dies **202(1)**, **202(2)** to the first side **214(1)** of the package substrate **206** and disposing the underfill material **230** adjacent to and underneath the dies **202(1)**, **202(2)** to form the IC package **500** (block **1104** in FIG. **11A**). Note that in this example, because the stiffener structure **208** is part of a monolithic structure of the heat spreader structure **508**, the dies **202(1)**, **202(2)** are coupled to the package substrate **206** before the heat spreader

structure **508** is placed and coupled to the package substrate **206** with its cavities **212(1)**, **212(2)** encompassing the first and second electrical components **204(1)**, **204(2)**. This is because the cover structure **510** of the heat spreader structure **508**, if coupled to the package substrate **206** before the dies **202(1)**, **202(2)**, would block access for the dies **202(1)**, **202(2)** to be coupled to the package substrate **206** and the underfill material **230** disposed around the dies **202(1)**, **202(2)**. Thus, in this example, the stiffener structure **208** cannot act as an underfill shield or dam from the underfill material **230**.

[0053] Then, as illustrated in fabrication stage **1200C** in FIG. **12C**, the fabrication process **1100** includes disposing the adhesive material **218** on the first side **214(1)** of the package substrate **206** where the second surface **224** of the stiffener structure **208** will be coupled, and disposing the thermal transfer material **402(1)**, **402(2)** on the first and second dies **202(1)**, **202(2)** (block **1106** in FIG. **11B**). This is to prepare the package substrate **206** to receive and be coupled to the heat spreader structure **508**. Then, as illustrated in fabrication stage **1200D** in FIG. **12D**, the fabrication process **1100** includes coupling the heat spreader structure **508** on the package substrate **206** such that second surface **224** of the stiffener structure **208** is coupled to the adhesive material **218** on the first side **214(1)** of the package substrate **206** and the cover structure **510** is coupled to the thermal transfer material **402(1)**, **402(2)** with the first and second electrical components **204(1)**, **204(2)** encompassed in the first and second cavities **212(1)**, **212(2)** of the stiffener structure **208** to form the IC package **500** (block **1108** in FIG. **11D**). In this regard, the heat spreader structure **508** is coupled to the package substrate **206** to add stiffness to the package substrate **206** while also providing a heat sink to dissipate heat generated by the first and second dies **202(1)**, **202(2)**.

[0054] It should be understood that the terms “first,” “second,” “third,” etc., where used herein, are relative terms and are not meant to limit or imply a strict orientation. It should also be understood that the terms “top,” “upper,” “above,” and “bottom,” “lower,” “below,” where used herein, are relative terms and are not meant to limit or imply a strict orientation. A “top” or “upper” or “above” referenced element does not always need to be oriented to be above a “bottom,” or “lower,” or “below” referenced element with respect to ground, and vice versa. An element referenced as “top,” “upper,” “above,” or “bottom,” “lower,” “below,” may be on top or bottom relative to that example only and the particular illustrated example. An element referenced as “top” or “upper” or “above” “bottom,” “lower,” “below,” another element does not have to be with respect to ground, and vice versa. An element referenced as “top” or “upper” or “above” may be above or below such other referenced element, relative to that example only and the particular illustrated example.

[0055] Further, an object being “adjacent” as discussed herein relates to an object being beside or next to another stated object. Adjacent objects may not be directly physically coupled to each other. An object can be directly adjacent to another object which means that such objects are directly beside or next to the other object without another object or layer being intervening or disposed between the directly adjacent objects. An object can be indirectly or non-directly adjacent to another object which means that such objects are not directly beside or directly next to each other, but there is an intervening object or layer disposed between the non-directly adjacent objects.

[0056] IC packages that include a stiffener structure coupled to a first side of a package substrate to increase stiffness of the IC package, and wherein the stiffener structure includes an integrated cavity(ies) extending to the first side of the package substrate and encompassing an electrical component(s) coupled to the first side of the package substrate, including, but not limited to, the IC packages **200**, **300**, **400**, **500** in FIGS. **2A-5B**, and that can be fabricated according to any fabrication process, including, but not limited to, the fabrication processes **600**, **700**, **900**, **1100** in FIGS. **6**, **7A-7B**, **9A-9C**, and **11A-11B**, and according to any aspects disclosed herein, may be provided in or integrated into any processor-based device. Examples, without limitation, include a set top box, an entertainment unit, a navigation device, a communications device, a fixed location data unit, a mobile location data unit, a global positioning system (GPS) device, a mobile phone, a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a tablet, a phablet, a server,

a computer, a portable computer, a mobile computing device, a wearable computing device (e.g., a smart watch, a health or fitness tracker, eyewear, etc.), a desktop computer, a personal digital assistant (PDA), a monitor, a computer monitor, a television, a tuner, a radio, a satellite radio, a music player, a digital music player, a portable music player, a digital video player, a video player, a digital video disc (DVD) player, a portable digital video player, an automobile, a vehicle component, avionics systems, a drone, and a multicopter.

[0057] In this regard, FIG. 13 illustrates an exemplary wireless communications device **1300** that includes electrical components formed from one or more ICs **1302**, wherein any of the ICs **1302** can be included in an IC package **1303**, **1303(1)**, **1303(2)**. The IC packages **1303**, **1303(1)**, **1303(2)** can include a stiffener structure coupled to a first side of a package substrate to increase stiffness of the IC package, and wherein the stiffener structure includes an integrated cavity(ies) extending to the first side of the package substrate and encompassing an electrical component(s) coupled to the first side of the package substrate, including, but not limited to, the IC packages **200**, **300**, **400**, **500** in FIGS. 2A-5B, and that can be fabricated according to any fabrication process, including, but not limited to, the fabrication processes **600**, **700**, **900**, **1100** in FIGS. 6, 7A-7B, 9A-9C, and 11A-11B, and according to any aspects disclosed herein. The wireless communications device **1300** may include or be provided in any of the above referenced devices, as examples. As shown in FIG. 13, the wireless communications device **1300** includes a transceiver **1304** and a data processor **1306**. The data processor **1306** may include a memory to store data and program codes. The transceiver **1304** includes a transmitter **1308** and a receiver **1310** that support bi-directional communications. In general, the wireless communications device **1300** may include any number of transmitters **1308** and/or receivers **1310** for any number of communication systems and frequency bands. All or a portion of the transceiver **1304** may be implemented on one or more analog ICs, RF ICs (RFICs), mixed-signal ICs, etc.

[0058] The transmitter **1308** or the receiver **1310** may be implemented with a super-heterodyne architecture or a direct-conversion architecture. In the super-heterodyne architecture, a signal is frequency-converted between RF and baseband in multiple stages, e.g., from RF to an intermediate frequency (IF) in one stage, and then from IF to baseband in another stage for the receiver **1310**. In the direct-conversion architecture, a signal is frequency-converted between RF and baseband in one stage. The super-heterodyne and direct-conversion architectures may use different circuit blocks and/or have different requirements. In the wireless communications device **1300** in FIG. 13, the transmitter **1308** and the receiver **1310** are implemented with the direct-conversion architecture.

[0059] In the transmit path, the data processor **1306** processes data to be transmitted and provides I and Q analog output signals to the transmitter **1308**. In the exemplary wireless communications device **1300**, the data processor **1306** includes digital-to-analog converters (DACs) **1312(1)**, **1312(2)** for converting digital signals generated by the data processor **1306** into the I and Q analog output signals, e.g., I and Q output currents, for further processing.

[0060] Within the transmitter **1308**, lowpass filters **1314(1)**, **1314(2)** filter the I and Q analog output signals, respectively, to remove undesired signals caused by the prior digital-to-analog conversion. Amplifiers (AMPs) **1316(1)**, **1316(2)** amplify the signals from the lowpass filters **1314(1)**, **1314(2)**, respectively, and provide I and Q baseband signals. An upconverter **1318** upconverts the I and Q baseband signals with I and Q transmit (TX) local oscillator (LO) signals through mixers **1320(1)**, **1320(2)** from a TX LO signal generator **1322** to provide an upconverted signal **1324**. A filter **1326** filters the upconverted signal **1324** to remove undesired signals caused by the frequency upconversion as well as noise in a receive frequency band. A power amplifier (PA) **1328** amplifies the upconverted signal **1324** from the filter **1326** to obtain the desired output power level and provides a transmit RF signal. The transmit RF signal is routed through a duplexer or switch **1330** and transmitted via an antenna **1332**.

[0061] In the receive path, the antenna **1332** receives signals transmitted by base stations and provides a received RF signal, which is routed through the duplexer or switch **1330** and provided to

a low noise amplifier (LNA) **1334**. The duplexer or switch **1330** is designed to operate with a specific receive (RX)-to-TX duplexer frequency separation, such that RX signals are isolated from TX signals. The received RF signal is amplified by the LNA **1334** and filtered by a filter **1336** to obtain a desired RF input signal. Downconversion mixers **1338(1)**, **1338(2)** mix the output of the filter **1336** with I and Q RX LO signals (i.e., LO_I and LO_Q) from an RX LO signal generator **1340** to generate I and Q baseband signals. The I and Q baseband signals are amplified by AMPs **1342(1)**, **1342(2)** and further filtered by lowpass filters **1344(1)**, **1344(2)** to obtain I and Q analog input signals, which are provided to the data processor **1306**. In this example, the data processor **1306** includes analog-to-digital converters (ADCs) **1346(1)**, **1346(2)** for converting the analog input signals into digital signals to be further processed by the data processor **1306**.

[0062] In the wireless communications device **1300** of FIG. **13**, the TX LO signal generator **1322** generates the I and Q TX LO signals used for frequency upconversion, while the RX LO signal generator **1340** generates the I and Q RX LO signals used for frequency downconversion. Each LO signal is a periodic signal with a particular fundamental frequency. A TX phase-locked loop (PLL) circuit **1348** receives timing information from the data processor **1306** and generates a control signal used to adjust the frequency and/or phase of the TX LO signals from the TX LO signal generator **1322**. Similarly, an RX PLL circuit **1350** receives timing information from the data processor **1306** and generates a control signal used to adjust the frequency and/or phase of the RX LO signals from the RX LO signal generator **1340**.

[0063] FIG. **14** illustrates an example of a processor-based system **1400** that can include one or more IC packages **1402**, **1402(1)**-**1402(8)** that include a stiffener structure coupled to a first side of a package substrate to increase stiffness of the IC package, and wherein the stiffener structure includes an integrated cavity(ies) extending to the first side of the package substrate and encompassing an electrical component(s) coupled to the first side of the package substrate, including, but not limited to, the IC packages **200**, **300**, **400**, **500** in FIGS. **2A-5B**, and that can be fabricated according to any fabrication process, including, but not limited to, the fabrication processes **600**, **700**, **900**, **1100** in FIGS. **6**, **7A-7B**, **9A-9C**, and **11A-11B**, and according to any aspects disclosed herein.

[0064] In this example, the processor-based system **1400** may be formed as an IC **1404** in an IC package **1402** and as a system-on-a-chip (SoC) **1406**. In this example, the processor-based system **1400** may be provided as or include a system-on-a-chip (SoC) **1406**. The processor-based system **1400** includes a CPU **1408** that includes one or more processors **1410**, which may also be referred to as CPU cores or processor cores. The CPU **1408** can be included in an IC package **1402(1)**. The CPU **1408** may have cache memory **1412** coupled to the CPU **1408** for rapid access to temporarily stored data. The CPU **1408** is coupled to a system bus **1414** and can intercouple master and slave devices included in the processor-based system **1400**. As is well known, the CPU **1408** communicates with these other devices by exchanging address, control, and data information over the system bus **1414**. For example, the CPU **1408** can communicate bus transaction requests to a memory controller **1416** as an example of a slave device. Although not illustrated in FIG. **14**, multiple system buses **1414** could be provided, wherein each system bus **1414** constitutes a different fabric.

[0065] Other master and slave devices can be connected to the system bus **1414**. As illustrated in FIG. **14**, these devices can include a memory system **1420** that includes the memory controller **1416** and a memory array(s) **1418**, one or more input devices **1422**, one or more output devices **1424**, one or more network interface devices **1426**, and one or more display controllers **1428**, as examples. The memory system **1420** can be included in an IC package **1402(2)**. The network interface devices **1426** can be included in an IC package **1402(3)**. Each of the memory system **1420**, the one or more input devices **1422**, the one or more output devices **1424**, the one or more network interface devices **1426**, and the one or more display controllers **1428** can be provided in the same or different circuit packages. The input devices **1422** and/or the output devices **1424** can

be included in respective IC packages **1402(4)**, **1402(5)** to dissipate heat. The input device(s) **1422** can include any type of input device, including, but not limited to, input keys, switches, voice processors, etc. The output device(s) **1424** can include any type of output device, including, but not limited to, audio, video, other visual indicators, etc. The network interface device(s) **1426** can be any device configured to allow exchange of data to and from a network **1430**. The network **1430** can be any type of network, including, but not limited to, a wired or wireless network, a private or public network, a local area network (LAN), a wireless local area network (WLAN), a wide area network (WAN), a BLUETOOTH™ network, and the Internet. The network interface device(s) **1426** can be configured to support any type of communications protocol desired.

[0066] The CPU **1408** may also be configured to access the display controller(s) **1428** over the system bus **1414** to control information sent to one or more displays **1432**. The display **1432** can be included in an IC package **1402(6)** to dissipate heat. The display controller(s) **1428** sends information to the display(s) **1432** to be displayed via one or more video processors **1434**, which process the information to be displayed into a format suitable for the display(s) **1432**. The display controller(s) **1428** and video processor(s) **1434** can be included in respective IC packages **1402(7)**, **1402(8)** or the same IC package, and/or in the same or different IC package **1402(1)** containing the CPU **1408** as an example. The display(s) **1432** can include any type of display, including, but not limited to, a cathode ray tube (CRT), a liquid crystal display (LCD), a plasma display, a light emitting diode (LED) display, etc.

[0067] Those of skill in the art will further appreciate that the various illustrative logical blocks, modules, circuits, and algorithms described in connection with the aspects disclosed herein may be implemented as electronic hardware, instructions stored in memory or in another computer readable medium and executed by a processor or other processing device, or combinations of both. The master and slave devices described herein may be employed in any circuit, hardware component, IC, or IC chip, as examples. Memory disclosed herein may be any type and size of memory and may be configured to store any type of information desired. To clearly illustrate this interchangeability, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. How such functionality is implemented depends upon the particular application, design choices, and/or design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

[0068] The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a processor, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

[0069] The aspects disclosed herein may be embodied in hardware and in instructions that are stored in hardware, and may reside, for example, in Random Access Memory (RAM), flash memory, Read Only Memory (ROM), Electrically Programmable ROM (EPROM), Electrically Erasable Programmable ROM (EEPROM), registers, a hard disk, a removable disk, a CD-ROM, or any other form of computer readable medium known in the art. An exemplary storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a remote

station. In the alternative, the processor and the storage medium may reside as discrete components in a remote station, base station, or server.

[0070] It is also noted that the operational steps described in any of the exemplary aspects herein are described to provide examples and discussion. The operations described may be performed in numerous different sequences other than the illustrated sequences. Furthermore, operations described in a single operational step may actually be performed in a number of different steps. Additionally, one or more operational steps discussed in the exemplary aspects may be combined. It is to be understood that the operational steps illustrated in the flowchart diagrams may be subject to numerous different modifications as will be readily apparent to one of skill in the art. Those of skill in the art will also understand that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0071] The previous description of the disclosure is provided to enable any person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations. Thus, the disclosure is not intended to be limited to the examples and designs described herein, but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

[0072] Implementation examples are also described in the following numbered clauses:

1. An integrated circuit (IC) package, comprising: [0073] a package substrate comprising a first side extending in a first direction; [0074] a first die coupled to the first side of the package substrate; [0075] a first stiffener structure adjacent to the first die in the first direction and coupled to the first side of the package substrate, the first stiffener structure comprising: [0076] a first cavity extending adjacent to the first side of the package substrate; and [0077] a first electrical component coupled to the first side of the package substrate, the first electrical component at least partially embedded in the first cavity.
2. The IC package of clause 1, wherein the first stiffener structure does not intersect the first die in a second direction orthogonal to the first direction.
3. The IC package of clause 1 or 2, wherein the first cavity does not intersect the first die in a second direction orthogonal to the first direction.
4. The IC package of any of clauses 1-3, wherein the first electrical component is fully embedded in the first cavity.
5. The IC package of any of clauses 1-4, wherein the first stiffener structure further comprises: [0078] a first surface adjacent to the package substrate; and [0079] a second surface opposite the first surface in a second direction orthogonal to the first direction; [0080] the first cavity extending through to the first surface of the first stiffener structure to extend adjacent to the first side of the package substrate.
6. The IC package of any of clauses 1-5, wherein the first stiffener structure further comprises: [0081] a first outer side wall adjacent to the first die; and [0082] a second outer side wall opposite the first outer side wall in the first direction; [0083] the first cavity between the first outer side wall and the second outer side wall in the first direction.
7. The IC package of clause 6, further comprising an underfill material adjacent to the first side of the package substrate and the first die; [0084] wherein the first outer side wall is adjacent to the underfill material.
8. The IC package of clause 7, wherein the first outer side wall is in contact with the underfill material.
9. The IC package of any of clauses 1-8, wherein the first electrical component is a first distance in a second direction orthogonal to the first direction between 0.65 and 1.2 millimeters (mm) from a

side wall of the first die adjacent to the first stiffener structure.

10. The IC package of clause 5, wherein the first cavity further extends to the second surface of the first stiffener structure.

11. The IC package of any of clauses 1-10, wherein the first stiffener structure comprises a ring-shaped stiffener structure.

12. The IC package of clause 11, wherein the ring-shaped stiffener structure is adjacent to a perimeter of the package substrate.

13. The IC package of any of clauses 1-12, wherein: [0085] the first stiffener structure further comprises a second cavity extending adjacent to the first side of the package substrate; and [0086] further comprising: [0087] a second electrical component coupled to the first side of the package substrate, the second electrical component at least partially embedded in the second cavity.

14. The IC package of clause 13, wherein the first cavity and the second cavity are part of a common cavity in the first stiffener structure.

15. The IC package of any of clauses 1-14, further comprising: [0088] a second stiffener structure outside the first die in the first direction and coupled to the first side of the package substrate, the second stiffener structure comprising: [0089] a second cavity extending adjacent to the first side of the package substrate; and [0090] a second electrical component coupled to the first side of the package substrate, the second electrical component at least partially embedded in the second cavity.

16. The IC package of any of clauses 1-15, wherein the first electrical component is electrically coupled to the first die through the package substrate.

17. The IC package of any of clauses 1-16, further comprising a heat spreader structure comprising the first stiffener structure and a cover structure extending in the first direction adjacent to the first die and fully intersecting the first die in a second direction orthogonal to the first direction.

18. The IC package of clause 17, wherein the first stiffener structure and the cover structure are a monolithic structure.

19. The IC package of any of clauses 1-18, [0091] wherein the first die comprises a first chiplet; and [0092] further comprising a second chiplet coupled to the first side of the package substrate; [0093] wherein: [0094] the first chiplet electrically coupled to the second chiplet through the package substrate; and [0095] the first stiffener structure is further adjacent to the second chiplet in the first direction.

20. The IC package of any of clauses 1-19 integrated into a device selected from the group consisting of: a set top box; an entertainment unit; a navigation device; a communications device; a fixed location data unit; a mobile location data unit; a global positioning system (GPS) device; a mobile phone; a cellular phone; a smart phone; a session initiation protocol (SIP) phone; a tablet; a phablet; a server; a computer; a portable computer; a mobile computing device; a wearable computing device; a desktop computer; a personal digital assistant (PDA); a monitor; a computer monitor; a television; a tuner; a radio; a satellite radio; a music player; a digital music player; a portable music player; a digital video player; a video player; a digital video disc (DVD) player; a portable digital video player; an automobile; a vehicle component; avionics systems; a drone; and a multicopter.

21. A method of fabricating an integrated circuit (IC) package, comprising: [0096] providing a package substrate comprising a first side extending in a first direction; [0097] coupling a first electrical component to the first side of the package substrate; [0098] coupling a first surface of a first stiffener structure comprising a first cavity extending adjacent to the first surface, to the first side of the package substrate such that the first cavity at least partially encloses the first electrical component; and [0099] coupling a first die to the first side of the package substrate adjacent to the first stiffener structure in the first direction.

22. The method of clause 21, wherein coupling the first surface of the first stiffener structure further comprises not intersecting the first stiffener structure with the first die in a second direction orthogonal to the first direction.

23. The method of clause 21 or 22, wherein coupling the first surface of the first stiffener structure further comprises the first cavity not intersecting the first die in a second direction orthogonal to the first direction.
24. The method of any of clauses 21-23, further comprising coupling a cover structure to the first stiffener structure extending in the first direction adjacent to the first die and fully intersecting the first die in a second direction orthogonal to the first direction.
25. The method of any of clauses 21-24, further comprising coupling the first electrical component to the first side of the package substrate prior to coupling the first surface of the first stiffener structure to the first side of the package substrate.
26. The method of any of clauses 21-25, further comprising disposing an underfill material adjacent to the first die after coupling the first surface of the first stiffener structure to the first side of the package substrate.

Claims

1. An integrated circuit (IC) package, comprising: a package substrate comprising a first side extending in a first direction; a first die coupled to the first side of the package substrate; a first stiffener structure adjacent to the first die in the first direction and coupled to the first side of the package substrate, the first stiffener structure comprising: a first cavity extending adjacent to the first side of the package substrate; and a first electrical component coupled to the first side of the package substrate, the first electrical component at least partially embedded in the first cavity.
2. The IC package of claim 1, wherein the first stiffener structure does not intersect the first die in a second direction orthogonal to the first direction.
3. The IC package of claim 1, wherein the first cavity does not intersect the first die in a second direction orthogonal to the first direction.
4. The IC package of claim 1, wherein the first electrical component is fully embedded in the first cavity.
5. The IC package of claim 1, wherein the first stiffener structure further comprises: a first surface adjacent to the package substrate; and a second surface opposite the first surface in a second direction orthogonal to the first direction; the first cavity extending through to the first surface of the first stiffener structure to extend adjacent to the first side of the package substrate.
6. The IC package of claim 1, wherein the first stiffener structure further comprises: a first outer side wall adjacent to the first die; and a second outer side wall opposite the first outer side wall in the first direction; the first cavity between the first outer side wall and the second outer side wall in the first direction.
7. The IC package of claim 6, further comprising an underfill material adjacent to the first side of the package substrate and the first die; wherein the first outer side wall is adjacent to the underfill material.
8. The IC package of claim 7, wherein the first outer side wall is in contact with the underfill material.
9. The IC package of claim 1, wherein the first electrical component is a first distance in a second direction orthogonal to the first direction between 0.65 and 1.2 millimeters (mm) from a side wall of the first die adjacent to the first stiffener structure.
10. The IC package of claim 5, wherein the first cavity further extends to the second surface of the first stiffener structure.
11. The IC package of claim 1, wherein the first stiffener structure comprises a ring-shaped stiffener structure.
12. The IC package of claim 11, wherein the ring-shaped stiffener structure is adjacent to a perimeter of the package substrate.
13. The IC package of claim 1, wherein: the first stiffener structure further comprises a second

cavity extending adjacent to the first side of the package substrate; and further comprising: a second electrical component coupled to the first side of the package substrate, the second electrical component at least partially embedded in the second cavity.

14. The IC package of claim 13, wherein the first cavity and the second cavity are part of a common cavity in the first stiffener structure.

15. The IC package of claim 1, further comprising: a second stiffener structure outside the first die in the first direction and coupled to the first side of the package substrate, the second stiffener structure comprising: a second cavity extending adjacent to the first side of the package substrate; and a second electrical component coupled to the first side of the package substrate, the second electrical component at least partially embedded in the second cavity.

16. The IC package of claim 1, wherein the first electrical component is electrically coupled to the first die through the package substrate.

17. The IC package of claim 1, further comprising a heat spreader structure comprising the first stiffener structure and a cover structure extending in the first direction adjacent to the first die and fully intersecting the first die in a second direction orthogonal to the first direction.

18. The IC package of claim 17, wherein the first stiffener structure and the cover structure are a monolithic structure.

19. The IC package of claim 1, wherein the first die comprises a first chiplet; and further comprising a second chiplet coupled to the first side of the package substrate; wherein: the first chiplet electrically coupled to the second chiplet through the package substrate; and the first stiffener structure is further adjacent to the second chiplet in the first direction.

20. The IC package of claim 1 integrated into a device selected from the group consisting of: a set top box; an entertainment unit; a navigation device; a communications device; a fixed location data unit; a mobile location data unit; a global positioning system (GPS) device; a mobile phone; a cellular phone; a smart phone; a session initiation protocol (SIP) phone; a tablet; a phablet; a server; a computer; a portable computer; a mobile computing device; a wearable computing device; a desktop computer; a personal digital assistant (PDA); a monitor; a computer monitor; a television; a tuner; a radio; a satellite radio; a music player; a digital music player; a portable music player; a digital video player; a video player; a digital video disc (DVD) player; a portable digital video player; an automobile; a vehicle component; avionics systems; a drone; and a multicopter.

21. A method of fabricating an integrated circuit (IC) package, comprising: providing a package substrate comprising a first side extending in a first direction; coupling a first electrical component to the first side of the package substrate; coupling a first surface of a first stiffener structure comprising a first cavity extending adjacent to the first surface, to the first side of the package substrate such that the first cavity at least partially encloses the first electrical component; and coupling a first die to the first side of the package substrate adjacent to the first stiffener structure in the first direction.

22. The method of claim 21, wherein coupling the first surface of the first stiffener structure further comprises not intersecting the first stiffener structure with the first die in a second direction orthogonal to the first direction.

23. The method of claim 21, wherein coupling the first surface of the first stiffener structure further comprises the first cavity not intersecting the first die in a second direction orthogonal to the first direction.

24. The method of claim 21, further comprising coupling a cover structure to the first stiffener structure extending in the first direction adjacent to the first die and fully intersecting the first die in a second direction orthogonal to the first direction.

25. The method of claim 21, further comprising coupling the first electrical component to the first side of the package substrate prior to coupling the first surface of the first stiffener structure to the first side of the package substrate.

26. The method of claim 21, further comprising disposing an underfill material adjacent to the first

die after coupling the first surface of the first stiffener structure to the first side of the package substrate.
